

IHP SG13G2 DRC Test Coverage Gap Analysis

Pre-existing issues exposed by PR #885 (Modular DRC Deck)

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1 Executive Summary

PR #885 ports the monolithic DRC deck (`sg13g2_maximal.drc`, 3071 lines, 272 rules) into a modular architecture of 49 files orchestrated by `ihp-sg13g2.drc`. This is the first time the modular orchestrator is activated in CI, exposing **170 pre-existing test framework gaps** that were previously hidden because the orchestrator had all include directives commented out.

Key findings:

- **0 DRC logic regressions:** cross-deck comparison shows 0 `COUNT_MISMATCH` across 40 unit testcases.
- **25 “Unknown” results:** caused by a parser bug in `run_regression.py`, not a DRC issue. Fix: 3 lines.
- **145 “Rule Not Tested”:** rules without dedicated testcase GDS files. 7 tables have no testcase at all; 6 have partial coverage.
- **Separate QA assets exist** (`libs.qa/drc/`) but are not integrated into the regression framework (Issue #925).

2 Background

2.1 The modular DRC deck (PR 885)

Item	Value
PR URL	https://github.com/IHP-GmbH/IHP-Open-PDK/pull/885
Branch	<code>feature/drc-modular-parity</code> on fork <code>Mauricio-xx/IHP-Open-PDK</code>
Base	<code>dev</code>
Commits	11 (signed-off, GPG-signed)
Rules ported	272/272 (100%)
Cross-deck parity	0 <code>COUNT_MISMATCH</code> on 40 unit GDS + 47 MB real chip layout
Runtime parity	maximal 41m08s vs modular 41m43s (mp=1)

2.2 Why CI never caught these issues before

On the `dev` branch, the modular orchestrator `ihp-sg13g2.drc` has **all 38 `%include` directives commented out**:

```
# File: ihp-sg13g2/libs.tech/klayout/tech/drc/ihp-sg13g2.drc (dev branch)
# %include rule_decks/layers_def.drc
# %include rule_decks/feol/5_1_nwell.drc
# %include rule_decks/feol/5_5_activ.drc
# ... (38 lines, all commented)
```

This means CI ran the orchestrator but **executed zero rules**. Combined with the GitHub Actions paths: filter (`ihp-sg13g2/libs.tech/klayout/tech/drc/**`), the `drc_regression` workflow only

triggers when a PR modifies DRC files. Recent PRs to `dev` (magic fixes, librelane, LVS updates) did not touch this path, so the workflow did not execute at all.

PR #885 activates the orchestrator (`#!/include` instead of `# %include`) and touches `drc/**`, triggering the workflow for the first time with actual rule execution.

2.3 CI exit code behavior

`run_regression.py` determines pass/fail at line 1125:

```
# File: testing/run_regression.py, line 1125
failing_results = df[~df["rule_status"].isin(["Passed"])]
```

Any rule status other than `Passed` causes exit 1. This includes `Rule Not Tested` and `Unknown`, not only `Rule Failed`. The Makefile target `test-DRC-main` wraps this, and GNU make converts the exit 1 into exit 2, which is what CI reports.

3 Analysis of the 25 “Unknown” Rules

3.1 Root cause: parser bug in RULES_VAR

All 25 `Unknown` results are in the `sealring` table. The function `parse_existing_rules()` in `run_regression.py` extracts rule names from `.output()` calls using regex, then expands Ruby `#{}` interpolations via a dictionary called `RULES_VAR` (lines 49–81).

Three variables are **missing** from `RULES_VAR`:

Variable	Used by	Expands to	Rules affected
<code>seal_a_name</code>	<code>Seal.a.#{seal_a_name}</code>	<code>Activ, pSD, Metal1–5, TopMetal1–2</code>	9
<code>cv_name</code>	<code>Seal.d.#{cv_name}</code>	<code>Cont, Via1–4, TopVia1–2</code>	7
<code>sf_name</code>	<code>Seal.f.#{sf_name}</code>	<code>Activ, pSD, Metal1–5, TopMetal1–2</code>	9

Without expansion, the parser sets `in_rule_deck=0` for these rules. Even though the DRC runs correctly (`viol_not_golden=0`, `golden_not_viol=0`), the status logic cannot assign `Passed` because `in_rule_deck` is false. The fallback status is `Unknown`.

3.2 Affected file and lines

File	Lines	What to change
<code>testing/run_regression.py</code>	49–81	Add 3 entries to <code>RULES_VAR</code> dict
<code>testing/gen_golden.py</code>	(verify)	Add same entries if <code>RULES_VAR</code> exists

3.3 Fix

```
# Add to RULES_VAR in testing/run_regression.py (lines 49–81):
"seal_a_name": ("Activ", "pSD", "Metal1", "Metal2", "Metal3",
```

```

        "Metal4", "Metal5", "TopMetal1", "TopMetal2"),
"cv_name": ("Cont", "Via1", "Via2", "Via3", "Via4",
            "TopVia1", "TopVia2"),
"sf_name": ("Activ", "pSD", "Metal1", "Metal2", "Metal3",
            "Metal4", "Metal5", "TopMetal1", "TopMetal2"),

```

3.4 DRC rule implementation (for reference)

The sealring rules are in `rule_decks/beol/6_10_sealring.drc`:

- **Seal.a** (lines 96–106): Min EdgeSeal-layer width = 3.50 um. Iterates over conductor layers.
- **Seal.d** (lines 179–196): Min EdgeSeal-Activ enclosure of contact/via ring = 1.30 um.
- **Seal.f** (lines 206–223): Min Passiv ring outside sealring space to EdgeSeal-layer.

All three use Ruby loops with interpolated output names. The DRC logic is correct; only the test parser fails to match the names.

4 Analysis of the 145 “Rule Not Tested”

4.1 Full inventory

4.1.1 Rules without any testcase GDS (7 tables, 103 rules)

offgrid (74 rules) – `rule_decks/offgrid/offgrid.drc`

All check 5nm grid alignment via `.ongrid()`. No `testing/testcases/unit/offgrid.gds` exists.

Rule	Layer	Rule	Layer
OffGrid.Activ	1:0	OffGrid.Activ_filler	1:22
OffGrid.Activ_nofill	1:23	OffGrid.Cont	6:0
OffGrid.DeepVia	–	OffGrid.DigiBnd	–
OffGrid.DigiSub	–	OffGrid.EXTBlock	–
OffGrid.EdgeSeal	–	OffGrid.EmWind	–
OffGrid.GatPoly	5:0	OffGrid.GatPoly_filler	5:22
OffGrid.GatPoly_nofill	5:23	OffGrid.IND	–
OffGrid.LBE	–	OffGrid.MIM	–
OffGrid.Metal1	8:0	OffGrid.Metal1_filler	8:22
OffGrid.Metal1_nofill	8:23	OffGrid.Metal1_slit	8:24
OffGrid.Metal2–5	10–49:0	(+ filler/nofill/slit variants)	
OffGrid.NWell	31:0	OffGrid.PWell	46:0
OffGrid.PWell_block	–	OffGrid.Passiv	–
OffGrid.TopMetal1–2	126–134:0	(+ filler/nofill/slit variants)	
OffGrid.TopVia1–2	–	OffGrid.Via1–4	–
OffGrid.pSD	14:0	OffGrid.nSD	7:0
OffGrid.nBuLay	32:0	OffGrid.nBuLay_block	–
OffGrid.nSD_block	–	OffGrid.SalBlock	–
OffGrid.ThickGateOx	–	OffGrid.TRANS	–
OffGrid.Vmim	–	OffGrid.SRAM	–
OffGrid.RFMEM	–	OffGrid.Recog_diode	–
OffGrid.Recog_esd	–	OffGrid.PolyRes	–
OffGrid.Polimide	–	OffGrid.NoMetFiller	–

Rule	Layer	Rule	Layer
OffGrid.dfpad	–	OffGrid.dfpad_pillar	–
OffGrid.dfpad_sbump	–		

rhig (6 rules) – rule_decks/feol/6_4_rhigh.drc

Rhi.a (width), Rhi.b (space), Rhi.c (enclosure), Rhi.d (overlap), Rhi.e (area), Rhi.f (density).
No rhig.gds exists.

rsil (6 rules) – rule_decks/feol/6_2_rsil.drc

Rsil.a–Rsil.f. No rsil.gds exists.

nmosi (5 rules) – rule_decks/feol/5_11_nmosi.drc

nmosi.b, nmosi.c, nmosi.d, nmosi.f, nmosi.g. No nmosi.gds exists.

rppd (5 rules) – rule_decks/feol/6_3_rppd.drc

Rppd.a–Rppd.e. No rppd.gds exists.

salblock (5 rules) – rule_decks/feol/5_12_salblock.drc

Sal.a–Sal.e. No salblock.gds exists.

extblock (3 rules) – location TBD

EXTB.a, EXTB.b, EXTB.c. No extblock.gds exists.

4.1.2 Rules with existing GDS but incomplete coverage (6 tables, 42 rules)

Table	Not Tested	Existing GDS	Missing rules
psd	21	psd.gds (21 KB)	pSD.a–n (17), nSDB.a/b/c/e (4)
metalslits	8	metalslits.gds (47 KB)	Slt.g.{M1–M4,TM2}, Slt.h2.{M1,TM1,TM2}
nbulay	4	nbulay_nBuLay.gds (16 KB)	NBLB.a/b/c/d
npnsubstratetie	3	npnsubstratetie.gds (35 MB)	npn13G2.a, npn13G2L.a, npn13G2V.a
nwell	3	nwell.gds (22 KB)	NW.c1.dig, NW.d1.dig, NW.e1.dig
sealring	2	sealring.gds (414 KB)	Seal.c1.TopVia1, Seal.c1.TopVia2

For **psd**: the GDS and golden exist but the golden was generated before the modular port added these rules. Regenerating the golden may resolve most or all 21 entries.

5 Relationship with Issue #925

Issue #925 (DRC test coverage: integrate `libs.qa` cells into regression framework) identifies a broader structural problem: the repository has **two separate sets of DRC test assets** that are not integrated.

5.1 The two test asset sets

Aspect	<code>libs.tech/.../testing/</code>	<code>libs.qa/drc/</code>
Purpose	Automated CI regression	Manual QA
Format	1 GDS per table + golden	Pass/fail pairs per rule
Coverage	48 GDS, 447 rules pass	31 cells, devices+layers
CI	Yes (<code>run_regression.py</code>)	None
Scripts	<code>gen_golden</code> , <code>run_regression</code>	None
Devices	No <code>rhigh</code> , <code>rppd</code> , <code>rsil</code> GDS	Has <code>rhigh</code> , <code>rppd</code> , <code>rsil</code>

5.2 The duplication problem

While creating testcase GDS files for `rhigh`, `rppd`, `rsil`, and other device tables, we built geometry from scratch. `libs.qa/drc/devices/` already had cells for several of these tables with proper device recognition stacks. The QA cells use a **pass/fail pair approach** that is arguably better for regression testing:

- Each rule has a “pass” cell (valid geometry, no violations expected) and a “fail” cell (intentionally violating geometry)
- The framework can validate both false positives (violations in pass cell) and false negatives (no violations in fail cell)

The current `run_regression.py` framework only uses golden-based comparison and expects one GDS per table with a specific naming convention. The QA cells cannot be directly consumed without adaptation.

5.3 Proposed resolution (from Issue #925)

Three options under discussion:

1. **Adapt the framework** to consume QA cells (add a mapping layer between QA cell names and the regression table structure)
2. **Restructure QA cells** to follow the `testing/testcases/unit/` naming convention
3. **Run both as separate CI suites** (parallel workflows)

This decision affects how we proceed with the 145 Rule Not Tested items. If the QA cells are integrated, many of the missing testcases are already solved.

6 Impact on CI

6.1 Current CI status for PR 885

Check	Status	Root cause
Python Code Linting	PASS	–
DCO	PASS	All 11 commits have Signed-off-by
drc_regression	FAIL	170 pre-existing non-Passed items (exit 2 from make)
drc_regression_cells	FAIL	Genuine pre-existing violations in SVaricap, pmos, pmosHV PCell templates

6.2 Projected fix effort

Phase	Items resolved	Effort	Dependency
RULES_VAR	25	Trivial (3 lines)	None
parser fix			
psd golden regeneration	up to 21	Low (scripted)	None
offgrid.gds creation	74	Low (scriptable)	None
6 new device GDS files	~30	Medium (manual KLayout)	Spec PDF
5 existing GDS enhancements	~16	Medium (manual KLayout)	Spec PDF
QA cell integration (Issue #925)	overlapping	High (framework changes)	Design decision

6.3 Recommended path forward

1. **Immediate:** fix RULES_VAR and regenerate psd golden (46 items, trivial effort).
2. **Short-term:** create offgrid.gds programmatically (74 items, scriptable).
3. **Medium-term:** decide on Issue #925 before creating remaining device GDS files. If QA cells are integrated, several tables are already covered.
4. **Do not relax run_regression.py exit semantics.** Fix the testcases rather than hiding the problem.

7 Appendix: Key File Paths

All paths relative to repository root (IHP-Open-PDK/).

7.1 DRC deck files

- `ihp-sg13g2/libs.tech/layout/tech/drc/ihp-sg13g2.drc`
Modular orchestrator (38 include directives)
- `ihp-sg13g2/libs.tech/layout/tech/drc/rule_decks/sg13g2_maximal.drc`
Monolithic reference deck (272 rules, 3071 lines)
- `ihp-sg13g2/libs.tech/layout/tech/drc/rule_decks/sg13g2_tech_default.json`
Default DRC parameter values

- `ihp-sg13g2/libs.tech/klayout/tech/drc/rule_decks/offgrid/offgrid.drc`
74 offgrid rules (single file)
- `ihp-sg13g2/libs.tech/klayout/tech/drc/rule_decks/beol/6_10_sealring.drc`
Sealring rules (Seal.a/d/f — the 25 Unknown rules)

7.2 Testing framework

- `ihp-sg13g2/libs.tech/klayout/tech/drc/testing/run_regression.py`
Golden regression script. RULES_VAR dict at lines 49–81.
- `ihp-sg13g2/libs.tech/klayout/tech/drc/testing/gen_golden.py`
Golden GDS generator
- `ihp-sg13g2/libs.tech/klayout/tech/drc/testing/run_crossdeck_comparison.py`
Cross-deck parity check (modular vs maximal)
- `ihp-sg13g2/libs.tech/klayout/tech/drc/testing/testcases/unit/`
41 testcase GDS files
- `ihp-sg13g2/libs.tech/klayout/tech/drc/testing/testcases/unit_golden/`
41 golden GDS files

7.3 QA assets (Issue #925)

- `ihp-sg13g2/libs.qa/drc/`
QA DRC cells (not integrated into CI)
- `ihp-sg13g2/libs.qa/drc/devices/`
Device-level pass/fail cells (rhigh, rppd, rsil, mim, sealring)

7.4 Spec documentation

- `ihp-sg13g2/libs.doc/doc/SG13G2_os_layout_rules.pdf`
Layout design rules specification
- `ihp-sg13g2/libs.doc/doc/SG13G2_os_process_spec.pdf`
Process specification

8 References

- PR #885 – Modular DRC deck
- PR #885 – CI status comment
- Issue #925 – DRC test coverage: integrate libs.qa cells
- PR #900 – Fix DRC JSON precedence
- PR #819 – DRC updates (added untested rules)